

Java Programming – Deep Dive Unit-wise Notes

This document provides deep conceptual explanations along with complete Java programs explained line by line. Designed for FYBSc Computer Science students for strong theory and programming clarity.

UNIT 1: Introduction to Java & Control Statements

Java Program: Hello World

This is the first basic Java program. Below is the complete program followed by line-by-line explanation.

```
class Hello {  
    public static void main(String[] args) {  
        System.out.println("Hello Java");  
    }  
}
```

Explanation:

class Hello → Defines a class named Hello.

public → Makes the method accessible from anywhere.

static → Allows JVM to call main method without creating object.

void → Specifies no return value.

main(String[] args) → Entry point of Java program.

System.out.println() → Prints output to console.

Program: Data Types and Variables

```
class DataTypes {  
    public static void main(String[] args) {  
        int a = 10;  
        float b = 5.5f;  
        char c = 'A';  
        boolean flag = true;  
  
        System.out.println(a);  
        System.out.println(b);  
        System.out.println(c);  
        System.out.println(flag);  
    }  
}
```

Each variable stores a specific type of data. int stores integers, float stores decimal values, char stores single character, and boolean stores true/false values.

Program: If-Else Statement

```
class IfElseDemo {  
    public static void main(String[] args) {
```

```

        int age = 18;

        if(age >= 18) {
            System.out.println("Eligible to vote");
        } else {
            System.out.println("Not eligible");
        }
    }
}

```

The if condition checks whether the condition is true. If true, the if block executes; otherwise, the else block executes.

Program: For Loop

```

class ForLoop {
    public static void main(String[] args) {
        for(int i = 1; i <= 5; i++) {
            System.out.println(i);
        }
    }
}

```

The for loop executes a block of code repeatedly until the condition becomes false.

Program: Array Example

```

class ArrayDemo {
    public static void main(String[] args) {
        int[] a = {10, 20, 30, 40};

        for(int i = 0; i < a.length; i++) {
            System.out.println(a[i]);
        }
    }
}

```

Arrays store multiple values of the same data type. The length property gives the size of the array.

Program: String Methods

```

class StringDemo {
    public static void main(String[] args) {
        String s = "Java Programming";

        System.out.println(s.length());
        System.out.println(s.toUpperCase());
        System.out.println(s.toLowerCase());
        System.out.println(s.substring(0, 4));
    }
}

```

Strings are objects in Java. The String class provides many built-in methods.

UNIT 2: Classes, Constructors, and Inheritance

Program: Class and Object

```
class Student {
    int id;
    String name;

    void display() {
        System.out.println(id + " " + name);
    }
}

class Main {
    public static void main(String[] args) {
        Student s1 = new Student();
        s1.id = 1;
        s1.name = "Tanmay";
        s1.display();
    }
}
```

The Student class defines data members and methods. The object s1 accesses class members using dot operator.

Program: Constructor

```
class Student {
    int id;
    String name;

    Student(int i, String n) {
        id = i;
        name = n;
    }

    void display() {
        System.out.println(id + " " + name);
    }
}

class Main {
    public static void main(String[] args) {
        Student s = new Student(1, "Amit");
        s.display();
    }
}
```

Constructors initialize objects. Parameterized constructor accepts values during object creation.

Program: Inheritance

```
class A {  
    void show() {  
        System.out.println("Class A");  
    }  
}  
  
class B extends A {  
    void display() {  
        System.out.println("Class B");  
    }  
}  
  
class Main {  
    public static void main(String[] args) {  
        B obj = new B();  
        obj.show();  
        obj.display();  
    }  
}
```

Inheritance allows class B to access properties of class A using the extends keyword.